

Computer network lab

1. **Lab work 1:** Study of different types of Networks cables and practically implements the cross-wired cable and straight through cable using clamping tool.
2. **Lab work 2:** Conduct a detailed study of the following networking devices, including their functions, types, advantages, and typical use cases:
 - a. Repeater
 - b. Hub
 - c. Switch
 - d. Bridge
 - e. Router
 - f. Gateway
 - g. Modem
 - h. Access Point (Wireless Access Point)
 - i. Network Interface Card (NIC)
 - j. Firewall
 - k. Proxy Server
 - l. Load Balancer
 - m. DNS Server
 - n. DHCP Server
3. **Lab work 3:** Study of basic network command and Network configuration commands.
 - a. ping
 - b. ipconfig
 - c. ifconfig
 - d. tracert
 - e. traceroute
 - f. netstat
 - g. nslookup
 - h. arp -a
 - i. hostname
 - j. route
 - k. ipconfig /release
 - l. ipconfig /renew
 - m. ipconfig /flushdns
 - n. netsh
 - o. telnet

- p. ftp
- q. ssh
- r. curl
- s. nmcli
- t. systemctl restart network

4. **Lab work 4:** Design and configure a small LAN using Packet Tracer with at least 4 PCs, a switch, and a router. Assign IP addresses manually and test connectivity using ping.
5. **Lab work 5:** Simulate and configure the following topologies using Packet Tracer:
 - a. Star
 - b. Ring
 - c. Bus
 - d. Mesh
 - e. Hybrid
6. **Lab work 6:** Bus and Star Topology Connection
Design a network in Cisco Packet Tracer that connects a Bus topology and a Star topology using a router. Use CLI to assign IP addresses manually to PCs and router interfaces. Test connectivity using ping between PCs on both topologies.
 - a. Draw the topology.
 - b. Assign IP addresses manually using CLI.
 - c. Show successful ping from a PC in the bus network to a PC in the star network.
7. **Lab Work 7:** Star and Tree Topology Connection
Implement a network topology that connects a Star and a Tree topology through a router. Use CLI mode to configure IP addresses on all devices.
 - a. Configure static IP addresses for all PCs and router interfaces.
 - b. Connect the two networks via the router.
 - c. Demonstrate connectivity between devices on different topologies using ping.
8. **Lab Work 8:** Star, Ring and Bus Topology Connection
Simulate the interconnection of Star, Ring and Bus topologies via a router in Cisco Packet Tracer. Use CLI to assign IP addresses to all PCs and router interfaces.
 - a. Use at least 3 PCs per topology.
 - b. Manually assign IP addresses and default gateways.

- c. Show screenshots or results of successful ping between PCs on separate topologies.
9. **Lab work 9:** Design and implement a network in Cisco Packet Tracer using at least 1 Router, 1 Switch, 1 server and 4 PCs. Configure DHCP server to automatically assign IP addresses to all PCs.
 10. **Lab work 10:** Simulate a network with two routers and configure RIP (Routing Information Protocol). Verify routing tables and connectivity. (2 Routers (e.g., Router0 and Router1), 2 Switches (one per LAN), 4 PCs (2 per LAN))
 11. **Lab work 11:** Simulate a network with three routers and configure RIP (Routing Information Protocol). Verify routing tables and connectivity. (3 Routers, 3 Switches (one per LAN), 6 PCs (2 per LAN))
 12. **Lab work 12:** Use FTP server in Packet Tracer to upload and download files between server and client.
 13. **Lab work 13:** In Cisco Packet Tracer, set up and configure a DNS server and a Web server. Host a sample HTML page on the Web server and access it from client PCs using a web browser via the domain name.
 14. **Lab work 14:** Configure a mail server and simulate sending/receiving emails between two users on different PCs.
 15. **Lab work 15:** Configure a basic firewall in Cisco Packet Tracer to control and filter traffic between two networks.

NOTE: Submission Instructions

1. All lab works must be **submitted in handwritten format**.
2. Prepare a lab report that includes a properly formatted cover page and an index (table of contents) page.
3. Your handwritten submission should include:
 - Detailed **steps for connection testing** (e.g., using ping or other network commands).
 - Clear **steps for IP address assignment** for devices.
 - Necessary **screenshots** showing the configuration and testing results.
4. Use the same format as followed in Lab Report 4 to prepare the remaining lab reports.